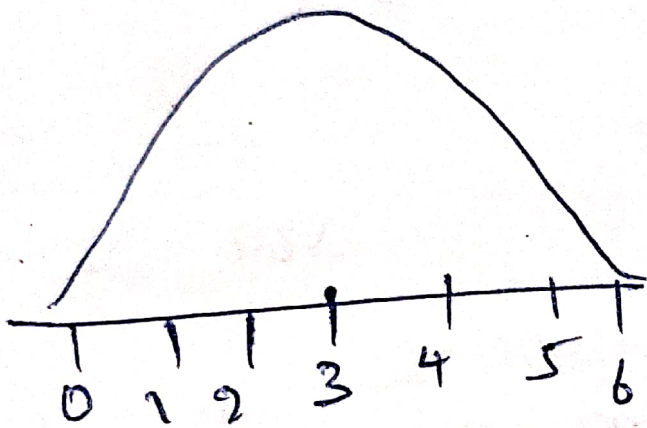


① Standard Normal distribution.

$$x = \{1, 2, 3, 4, 5\} \quad \mu = 3$$

$$\sigma = 1.14 \approx 1$$



$$\Rightarrow \mu = 0$$

$$\sigma = 1$$

$$x = \{1, 2, 3, 4, 5\}$$

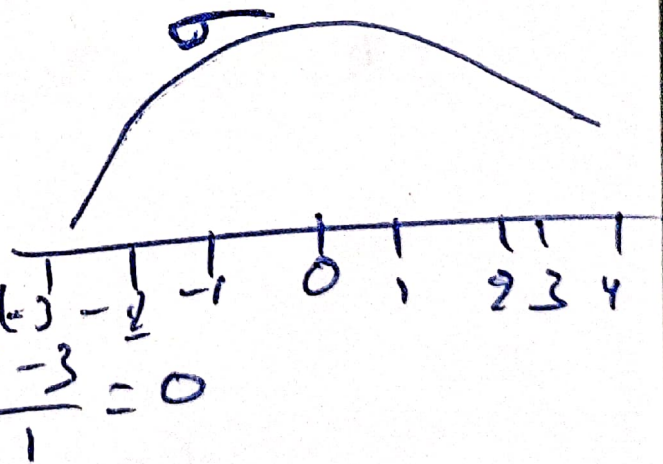
$$Z\text{-Score} = \frac{x_i - \mu}{\sigma}$$

$$y = \{-2, -1, 0, 1, 2\}$$

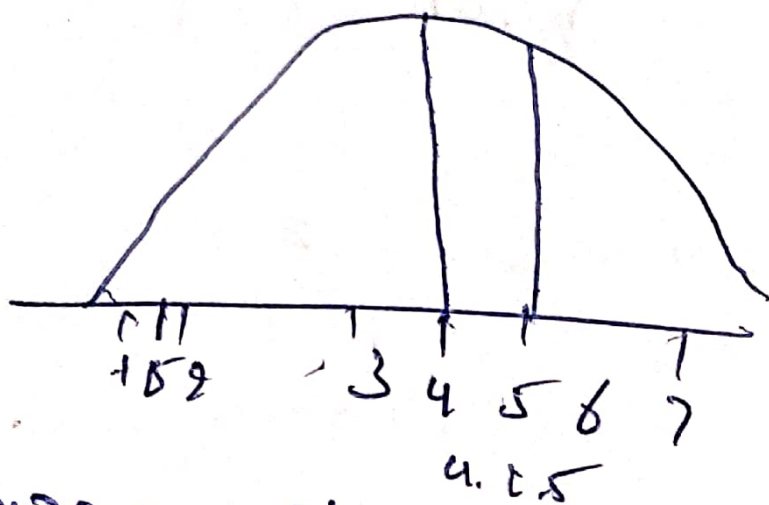
$$\textcircled{1} \quad \frac{1-3}{2} = -2$$

$$\textcircled{3} \quad \frac{3-3}{1} = 0$$

$$\textcircled{2} \quad \frac{2-3}{1} = -1$$



$$X \sim \text{N}(0, \sigma^2 = 1)$$



$$\mu = 4$$

$$\sigma = 1$$

How many standard deviations away from the mean.

$$z\text{-score} = \frac{4.25 - 4}{1}$$

$$x_i = 4.25$$

$$= 0.25$$

$$z_i = 2.5$$

$$= \frac{4.25 - 1.75}{1} = 2.5$$

Eg: Dataset

| Age | weight | height | sex  |
|-----|--------|--------|------|
| 24  | 70     | 175    | 40 K |
| 25  | 60     | 160    | 50 K |
| 26  | 55     | 150    | 60 K |
| 27  | 40     | 130    | 30 K |
| 30  | 30     | 105    | 20 K |
| 31  | 25     | 180    | 70 K |

1 Standardization.